Copilot

The **Tower of Hanoi** is a classic mathematical puzzle that challenges your problem-solving skills. Here's how it works:

1. **Setup**:
   - You have three rods (usually labeled A, B, and C) and a set of disks of different sizes.
   - The disks are stacked on one rod in decreasing order of size, with the largest disk at the bottom and the smallest at the top.
2. **Objective**:
   - The goal is to move all the disks from the starting rod to another rod (usually the target rod), following specific rules.
3. **Rules**:
   - Only one disk can be moved at a time.
   - You can only move the top disk from one stack to the top of another stack or to an empty rod.
   - Larger disks cannot be placed on top of smaller disks.
4. **Minimum Moves**:
   - The challenge lies in finding the **minimum number of moves** required to transfer the entire stack of disks from the source rod to the target rod.
5. **Recursive Solution**:
   - The Tower of Hanoi problem can be elegantly solved using recursion.
   - To move `n` disks from the source rod to the target rod:
     - Move the top `n-1` disks from the source rod to an auxiliary rod.
     - Move the largest disk (the `n`th disk) from the source rod to the target rod.
     - Finally, move the `n-1` disks from the auxiliary rod to the target rod.
6. **Example**:
   - Suppose you have 3 disks (labeled 1, 2, and 3) on rod A.
   - The minimum number of moves to transfer them to rod C following the rules is 7.

The Tower of Hanoi problem has applications in computer science, algorithm design, and teaching recursion. It's a delightful brain teaser! 🧠💡